Residual-shape inference and external-proxy audit of structural disturbance in SPARC rotation curves

Jozsef Olcsak

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Abstract

We ask whether fixed rotation-curve residual-shape features can recover externally reviewed structural-disturbance labels in SPARC galaxies, and whether independent H I disturbance proxies support the same diagnostic direction. The study reverses the Paper 1 audit: the A/C labels are treated as frozen external targets, while residual features are evaluated as predictors under leave-one-galaxy-out thresholding, shuffled-label null tests, bootstrap uncertainty, baseline-family controls, and observability stress checks. The primary internal feature, Projection RMS, reaches LOOGO AUC=0.771008403 with shuffled-label $p = 0.002000000$ and bootstrap 95% AUC interval $[0.600802469, 0.909100262]$. MOND-simple and empirical RAR-like residual scores also separate A/C systems, whereas a Newtonian baryonic RMS control is near chance, indicating a low-acceleration residual-family effect rather than projection-formula uniqueness. External proxy readouts are supportive but not decisive. The result is therefore promising but sample-limited: a reproducible residual-inference and external-proxy audit, not a Tau Core validation claim, not gravity-model selection, and not independent paper-grade external validation.

1 Introduction

Rotation-curve residuals are usually discussed as model error, but their radial structure can also carry information about non-equilibrium dynamics, non-circular motions, pressure support, beam smearing, inclination, and source-dependent observability [8, 6]. Paper 1 established a residual-blind association between externally assigned structural-disturbance labels and low-acceleration residual scatter in SPARC. Here we ask the inverse diagnostic question: can fixed residual-shape features recover those external labels better than chance?

The scope is deliberately narrow. This paper does not use residuals to redefine the labels, does not validate Tau Core, and does not select a unique gravity law. It tests whether a frozen residual feature map contains recoverable information about externally reviewed A/C disturbance classes, then checks whether independent H I disturbance proxies point in the same direction.

2 Data and frozen inputs

The working A/C sample contains 45 SPARC galaxies inherited from the Paper 1 reproducibility packet: 17 externally regular A systems and 28 externally disturbed C systems. SPARC rotation-curve and mass-model context follows [3, 2]. B-class systems remain excluded from the primary target labels because they encode uncertainty by construction. The residual features are computed from the fixed Paper 1 point map and are not retrained on the external-proxy catalogues.

The public packet contains the derived tables, scripts, figures, and audit records needed to regenerate the diagnostic results. Raw survey products and raw SPARC rotmod files are not redistributed by this repository.

The projection-family score is treated operationally as a fixed residual map inherited from Paper 1, without requiring the physical correctness of the underlying projection ansatz. This convention is central to the audit: the paper tests residual information content, not the physical validity of the projection model.

3 Residual-shape endpoint and validation design

For each galaxy g and radial point i, the packet starts from absolute log-residual magnitudes

$$r_{m,gi} = \left|\log V_{\mathrm{obs},gi} - \log V_{m,gi}\right|,$$

where m denotes the fixed residual family. Projection RMS is then

$$\mathrm{RMS}_{m,g} = \sqrt{\frac{1}{N_g}\sum_i r_{m,gi}^2}.$$

All points are weighted equally; no velocity-error weighting is applied in the primary endpoint. The radial domain is the available SPARC rotation-curve support for each galaxy after the inherited Paper 1 point-map construction. The same aggregation is used for Projection RMS, MOND-simple RMS, RAR-like RMS, and Newtonian baryonic RMS. The Projection endpoint is used as an operational residual-shape score rather than as a physical proof of the projection formula.

The primary internal validation uses leave-one-galaxy-out thresholding. For held-out galaxy h, the threshold is recomputed from the remaining galaxies:

$$T_h = \frac{1}{2}\left[\mathrm{median}(\mathrm{RMS}_{g\in A, g\neq h}) + \mathrm{median}(\mathrm{RMS}_{g\in C, g\neq h})\right].$$

The held-out galaxy is classified as C-like if its score exceeds T_h. A shuffled-label null preserves the 17/28 A/C class counts and uses 1000 shuffles with random seed 20260515; the practical p-value floor is therefore of order 10^{-3}, and the observed empirical value is $p = 0.002000000$. Bootstrap uncertainty uses 2000 class-stratified galaxy resamples with random seed 20260516 for the primary endpoint. These tests are internal to SPARC and should not be described as independent validation.

4 Primary results

Table 1: Primary residual-inference results.

Quantity	Value	Interpretation	Claim boundary
Projection RMS LOOGO AUC	0.771008403	primary diagnostic	SPARC-internal
Projection RMS shuffled-label p	0.002000000	label-null check	not replication
MOND-simple RMS AUC	0.720588235	low-acceleration family	not unique
RAR-like RMS AUC	0.731092437	low-acceleration family	not model selection
Newtonian baryonic RMS AUC	0.506302521	near-chance control	limited inference

Table 2: Bootstrap AUC intervals for baseline residual-score families.

Score family	AUC	CI low	CI high
Projection RMS	0.771008403	0.609243697	0.913865546
MOND-simple RMS	0.720588235	0.552521008	0.871848739
RAR-like RMS	0.731092437	0.569275210	0.878151261
Newtonian baryonic RMS	0.506302521	0.340336134	0.678571429

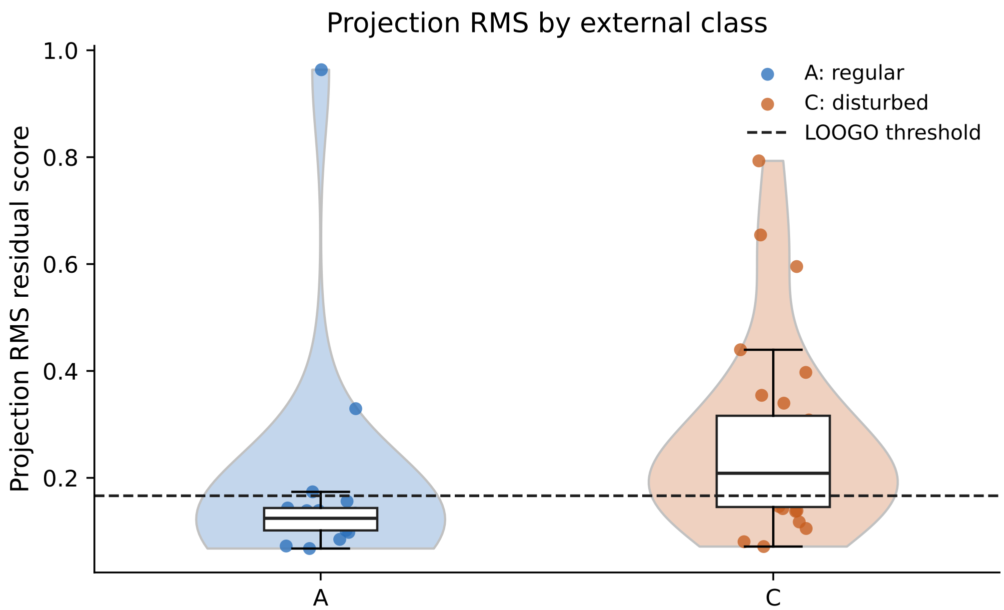

Figure 1: Distribution of the fixed Projection RMS residual score for externally regular A and disturbed C systems.

The primary signal is positive, but the bootstrap interval is broad. The correct reading is promising but sample-limited class recovery from residual shape, not a physical detection of an underlying field.

5 Baseline-family comparison

The baseline comparison weakens any projection-specific claim but strengthens the broader phenomenological result. MOND-simple [5] and empirical RAR-like [4] residual scores also separate A/C systems, while the Newtonian baryonic RMS score is near chance. The defensible conclusion is that A/C separation is concentrated in low-acceleration residual-family scores.

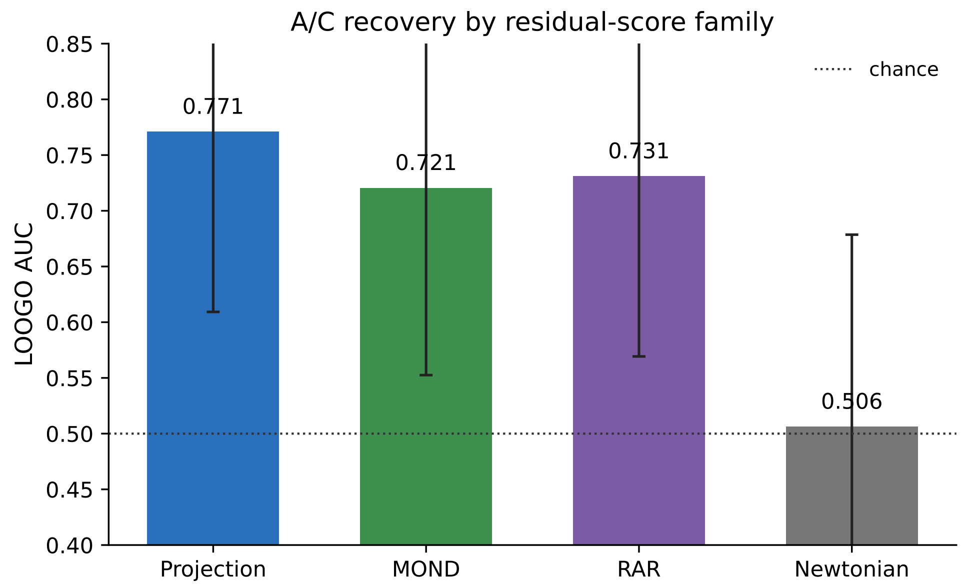

Figure 2: Held-out A/C recovery across residual-score families. The Newtonian baryonic RMS score is near chance.

This pattern is compatible with several physical readings: disturbed systems may violate smooth equilibrium assumptions, non-circular motions may be more important in low-acceleration regions, or observability may expose more disturbance structure in particular subsets. The present data do not distinguish these explanations.

6 Failure modes

The error audit is retained because a useful diagnostic must show where it fails. False-negative C systems demonstrate that externally disturbed galaxies do not always produce large residual burden under a smooth rotation-curve score. False-positive A systems show that residual structure can arise without a strong external disturbance label.

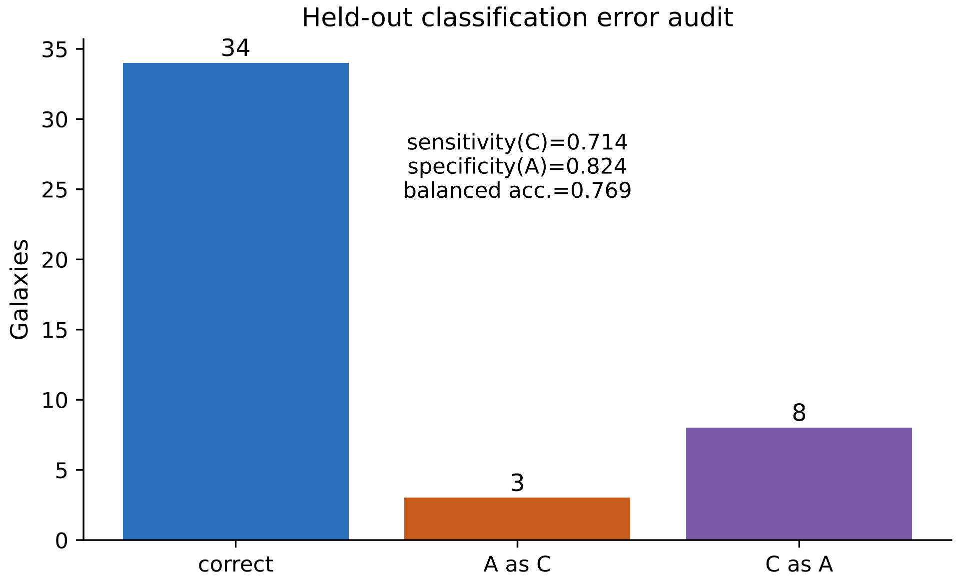

Figure 3: Held-out classification outcome counts for the Projection RMS threshold rule.

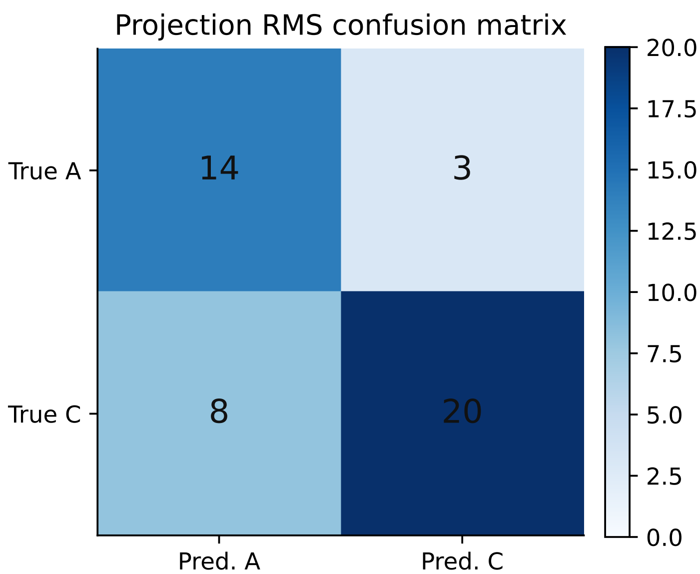

Figure 4: Confusion matrix for the held-out Projection RMS rule.

These failures are not relabeling evidence. They are targets for follow-up inspection, especially where residual morphology, inclination, radial coverage, and H I kinematic asymmetry disagree.

CamB is the most important failure case. It is externally labeled A from regular-kinematics evidence, but it has Projection RMS near 0.96 and only nine rotation-curve points. The paper therefore treats CamB as a named residual-hard outlier rather than letting it disappear inside summary statistics.

Table 3: B-class sensitivity check. B systems are not used as primary labels.

N_B	Threshold	B C-like	B A-like	Interpretation
28	0.166489797	13	15	descriptive only

Table 4: Named outliers and failure-case inspection.

Galaxy	Class	Error	Score	Points	Distance	Incl.	Inspection note
CamB	A	A as C	0.963585384	9	3.36	65.0	dominant A-class failure; low point count; very high low-acceleration residual burden
UGC05764	A	A as C	0.329468364	10	7.47	60.0	secondary A-class failure; compact 10-point curve; high residual margin
NGC5585	A	A as C	0.173858350	24	7.06	51.0	near-threshold A-class failure; external low-asymmetry evidence but residual just above gate

7 External proxy readouts

The external H I context draws on WHISP lopsidedness and morphology [9, 10, 1], THINGS non-circular-motion diagnostics [8], Reynolds et al. H I asymmetry catalogues [7], and ALFALFA profile asymmetry [11]. Table 5 lists each frozen proxy gate. WHISP resolved-HI asymmetry is the strongest supportive external readout, but its overlap is small. WHISP morphology is mixed, Reynolds/LVHIS is promising but below the frozen $N \geq 15$ gate, and ALFALFA/HALOGAS do not provide strong directional support. The external evidence therefore supports the paper as an audit, not as an independent validation result.

The strongest remaining weakness is external validation. All rotation-curve residual endpoints in this manuscript are SPARC-internal, and the auxiliary H I proxy readouts are contextual rather than a held-out replication. A fair skeptical reading is therefore: the association is reproducible within this SPARC packet, but it is not yet externally established.

Table 5: External-proxy readouts and frozen gates.

Source	Overlap	Direction/status	Effect size	Gate
WHISP resolved	14	positive, small N	r=0.391; AUC=0.714	context
WHISP morph.	25	mixed	AUC=0.644; burden=0.506	context
Reynolds/LVHIS	6	promising, N<15	r=0.375; AUC=0.778	fail N
ALFALFA	22	weak	r=0.146; AUC=0.472	weak
HALOGAS	5	control-like	r=0.216; AUC=0.500	control
THINGS route2	0_score_ready	closed	0 score-ready	negative audit

8 External-validation status

The present manuscript should not be read as a completed external-validation study. It contains three weaker ingredients: an internal SPARC residual-inference result, small-overlap external-proxy consistency checks, and a negative THINGS expansion audit. These ingredients are useful because they define the next test, but they do not replace that test.

A paper-grade Phase II replication must be a source-family holdout. The evidence rule, disturbance labels, observability covariates, minimum overlap, and pass/fail thresholds must be frozen before any endpoint readout. The minimum acceptable target remains $N \geq 15$ score-ready galaxies in a non-SPARC source family or in a demonstrably independent kinematic-proxy family. If the effect disappears, reverses, or becomes dominated by distance, inclination, point count, or beam-size proxies, the correct conclusion is that the current result is SPARC-specific, proxy-specific, or observability-driven.

9 THINGS route2 negative audit

The THINGS expansion route was audited because it could have raised the independent-control overlap. It is now closed as not score-ready. The required machine-readable per-radius mass-model columns were not recovered for the missing galaxies, and the reconstruction route failed its photometry and solver-validation gates before any missing-galaxy scores were computed.

This negative audit should remain visible. It documents why the paper does not claim THINGS $N \geq 15$, avoids plot-digitized or synthetic mass models, and prevents endpoint-driven data recovery from entering the evidence chain.

10 Observability and systematics

Observability remains the main scientific caveat. Nearby galaxies can reveal asymmetry and morphological disturbance more easily than distant systems; inclination, beam smearing, asymmetric drift, pressure support, and non-circular motion can all alter residual morphology. The current controls reduce but do not eliminate these concerns.

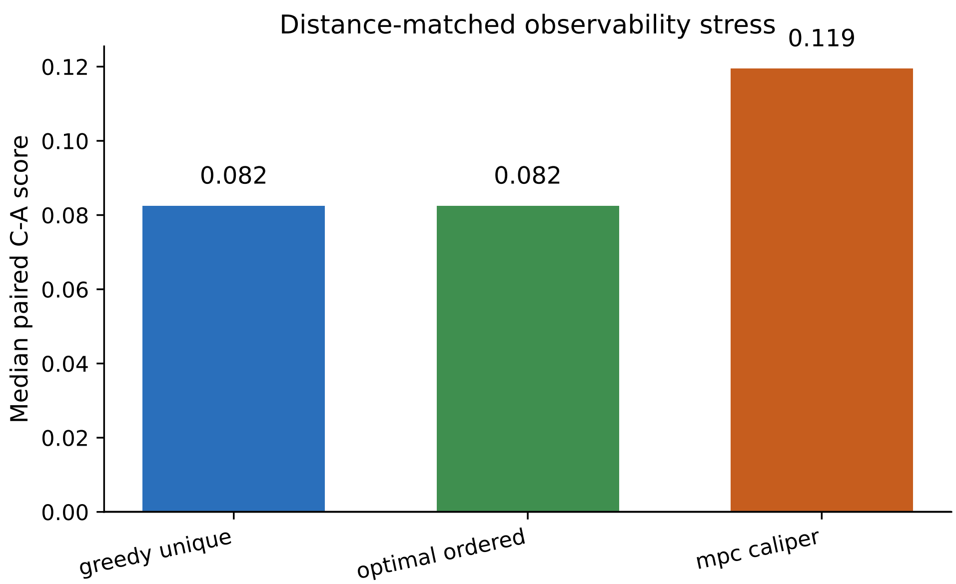

Figure 5: Distance-matched stress checks for the primary residual score. These checks reduce but do not eliminate observability concerns.

Accordingly, the result should be read as a reproducible SPARC diagnostic association with external-proxy context, not as a selection-function-proof physical inference.

The distance-stress labels are operational. Greedy unique matching pairs each A galaxy with one nearby C galaxy without reusing controls. Optimal ordered matching sorts both classes by distance and pairs by rank. The Mpc-caliper check keeps only pairs within the frozen maximum distance separation. These checks are not a full selection-function model.

Table 6: Observability and nuisance-covariate stress summaries.

Check	N	Value	Secondary value
partial distance residual correlation	45	0.434416486	nuisance controls
partial distance residual AUC	45	0.662000000	residual split
nuisance-only model R2	45	0.207747757	nuisance-only
point-count correlation	45	-0.256276232	nuisance channel
velocity-error correlation	45	-0.196321724	nuisance channel
inclination-error correlation	45	0.251685097	nuisance channel
distance-error correlation	45	-0.155582292	nuisance channel

This appendix-style stress table is not a full hierarchical model or a claim of bias removal. It documents that observability channels were inspected and that residual association should remain caveated until a larger external holdout can support multivariable inference.

11 Claim boundary and Phase II

Allowed claim: fixed residual-shape features recover externally reviewed A/C disturbance class better than chance in the current SPARC packet, and several external proxy readouts provide mixed but informative context.

Forbidden claims: Tau Core validation, gravity-model selection, projection-formula uniqueness, replacement of external labels by residual-only labels, broad independent external validation, or THINGS route2 positive evidence.

The next paper-grade step is a held-out external source-family test with $N \geq 15$, a frozen evidence rule, no velocity-endpoint refit, explicit observability covariates, and predefined failure conditions. A negative Phase II result should be treated as evidence that the present association is SPARC-specific, proxy-specific, or observability-driven.

12 Conclusion

This paper finds that fixed residual-shape features recover externally reviewed A/C disturbance classes better than chance within the current SPARC packet. The strongest internal endpoint is Projection RMS, with LOOGO AUC=0.771008403. MOND-simple and empirical RAR-like residual scores also separate the classes, while the Newtonian baryonic RMS control is near chance. The result points to a low-acceleration residual-family association with structural disturbance, not to a projection-formula-specific detection.

The paper does not establish a new gravity model, does not validate Tau Core, and does not replace external disturbance labels with residual-only labels. It also does not provide completed external validation. CamB and the other named failure cases show that residual burden can be high in externally regular systems, and the observability checks show that distance, radial coverage, inclination, point count, and H I data quality remain live alternative explanations.

The next decisive test is a held-out external source-family replication with a frozen evidence rule, at least $N \geq 15$ score-ready galaxies, observability covariates, and predefined failure conditions. If that test supports the same residual-disturbance direction, the present result becomes a stronger phenomenological claim. If it fails, the appropriate interpretation is that the association is SPARC-specific, proxy-specific, or observability-driven.

13 ROC appendix

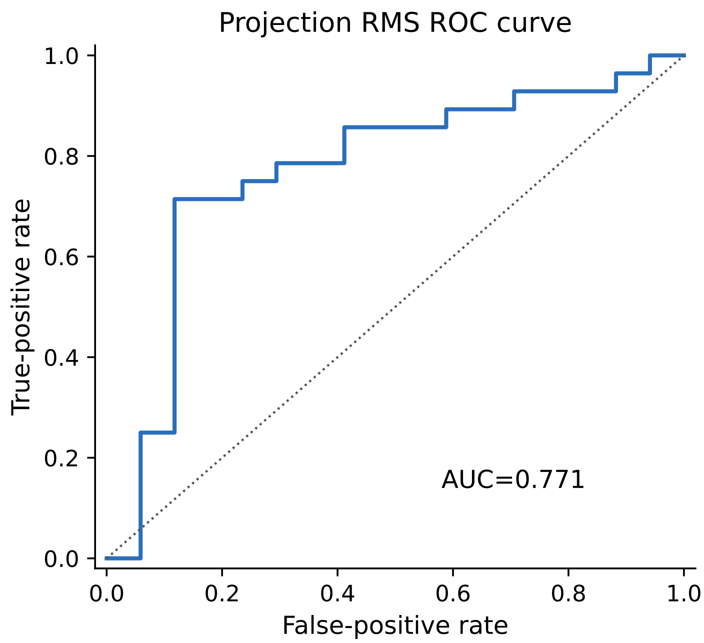

Figure 6: ROC curve for the Projection RMS score using the frozen A/C labels.

14 Stability and effect-size appendix

Table 7: Rank and stability effect-size summaries for Projection RMS.

Metric	Value	Secondary value
Mann–Whitney U	367.000000000	nA=17;nC=28
Cliff's δ	0.542016807	positive means C scores tend higher
AUC minus chance	0.271008403	auc=0.771008403
Bootstrap AUC median	0.775210084	p05=0.630252101;p95=0.894957983
Label-shuffle AUC median	0.497899160	p95=0.634453782;observed=0.771008403
AUC without CamB	0.819196429	A-class outlier removed for sensitivity only

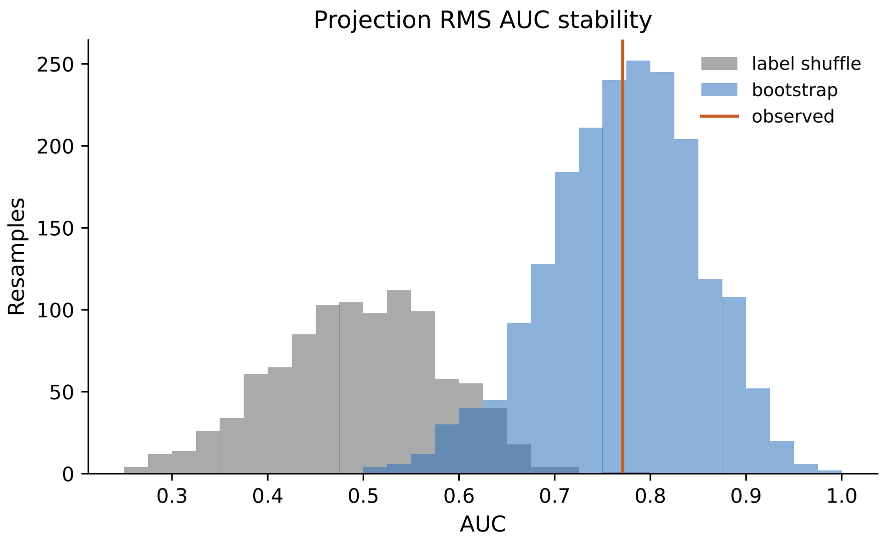

Figure 7: Bootstrap and shuffled-label AUC distributions for the Projection RMS endpoint.

A A/C sample table

Table 8: Frozen A/C sample used for the primary diagnostic.

Galaxy	Class	Distance	Incl.	Points	Projection RMS
CamB	A	3.36	65.0	9	0.963585384
DDO154	A	4.04	64.0	12	0.138812041
ESO079-G014	A	28.70	79.0	15	0.116776650
ESO116-G012	A	13.00	74.0	15	0.118483994
IC2574	C	3.91	75.0	34	0.221285260
NGC0055	C	2.11	77.0	21	0.204516651
NGC1705	C	5.73	80.0	14	0.354417125
NGC2366	C	3.27	68.0	26	0.250878781
NGC2403	A	3.16	63.0	73	0.084927422
NGC2976	C	3.58	61.0	27	0.239111203
NGC3109	C	1.33	70.0	25	0.175872342
NGC3198	A	13.80	73.0	43	0.156224718
NGC3741	C	3.21	70.0	21	0.204417844
NGC3917	A	18.00	79.0	17	0.143599282
NGC3972	A	18.00	77.0	10	0.101666153
NGC4183	A	18.00	82.0	23	0.067793794
NGC4559	A	9.00	67.0	32	0.133934337
NGC5055	C	9.90	55.0	28	0.117536453
NGC5585	A	7.06	51.0	24	0.173858350
NGC6503	A	6.26	74.0	31	0.072498887
NGC7331	A	14.70	75.0	36	0.098042642
UGC00128	A	64.50	57.0	22	0.138644763
UGC00191	A	17.10	45.0	9	0.124355499

Galaxy	Class	Distance	Incl.	Points	Projection RMS
UGC00731	C	12.50	57.0	12	0.339703437
UGC02455	C	6.92	51.0	8	0.793075490
UGC03580	C	20.70	63.0	47	0.280295930
UGC04305	C	3.45	40.0	22	0.595364038
UGC04325	C	9.60	41.0	8	0.284250584
UGC04499	C	12.50	50.0	9	0.137445134
UGC05253	C	22.90	37.0	73	0.080485995
UGC05716	A	21.30	54.0	12	0.113234582
UGC05764	A	7.47	60.0	10	0.329468364
UGC05829	C	8.64	34.0	11	0.138511849
UGC05918	C	7.66	46.0	8	0.142410680
UGC06446	C	12.00	51.0	17	0.194477890
UGC06818	C	18.00	75.0	8	0.308337913
UGC06917	C	18.00	56.0	11	0.071490152
UGC06973	C	18.00	71.0	9	0.212731540
UGC07323	C	8.00	47.0	10	0.146477589
UGC07399	C	8.43	55.0	10	0.397297780
UGC07577	C	2.59	63.0	9	0.654467132
UGC07603	C	4.70	78.0	12	0.180772538
UGC08490	C	4.65	50.0	30	0.186480775
UGC08837	C	7.21	80.0	8	0.439660106
UGC12632	C	9.77	46.0	15	0.105256872

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